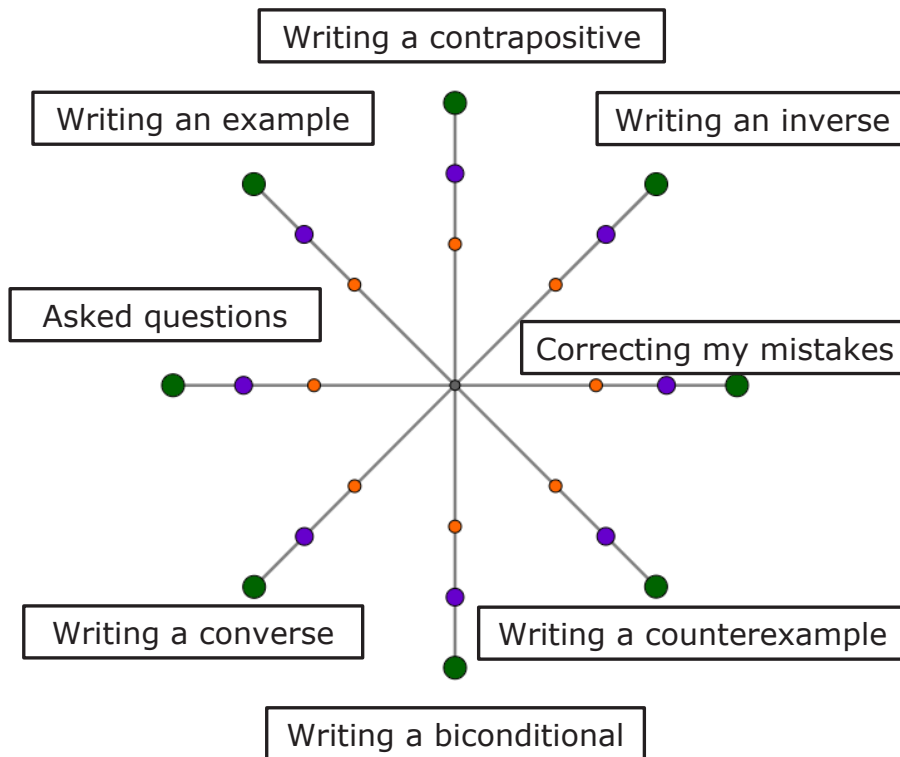




2.5.5 Wrap-Up: Reflection

- Place an **X** on the circle that shows what you learned or completed during today's lesson.

	I am confident I know how to do this!	OR	I always did this!
	I am fairly sure I know how to do this.	OR	I tried this.
	I need support on this.	OR	I had trouble with this.



Unit 2, Lesson 6: "All" and "Not" Statements



Benchmark

MA.912.LT.4.3 Identify and accurately interpret "if...then," "if and only if," "all" and "not" statements. Find the converse, inverse, and contrapositive of a statement.



Learning Targets

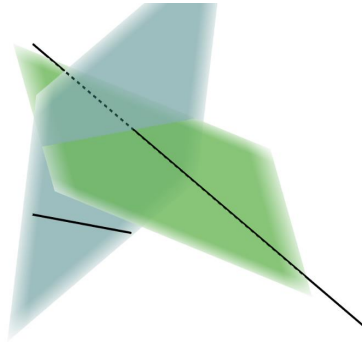
- ☐ I can write and interpret an "all" statement.
- ☐ I can write and interpret a negation.



2.6.1 Warm-Up: Bell Work

1. Complete the table for the conditional statement.

If two lines are skew, then the lines do not lie on the same plane.



Relationship to Conditional	Statement	True or False
Converse		
Inverse		
Contrapositive		

2. If the conditional statement is biconditional, then write it as a biconditional. If the conditional statement is not biconditional, then explain why not.



2.6.2 Guided Instruction: "All" and "Not" Statements

Mathematical statements may not be conditional or biconditional. Mathematical statements could be written using "all." The statement must be universally true to use the word "all."

1. Discuss with your partner if the statement, "All quadrilaterals have four sides" is universally true. Summarize your discussion.
2. Rewrite "All quadrilaterals have four sides" as an if-then statement.
3. Discuss with your partner why the statement "Some quadrilaterals have four sides" is not true. Summarize your discussion.

4. Determine if each statement is true.

Statement	True or False
All dogs are Labradors.	
All polygons have congruent sides.	
All cats are mammals.	

5. Rewrite each false statement using "some."

6. Write each statement as an if-then statement.

Statement	If-Then
All dogs are Labradors.	
All polygons have congruent sides.	
All cats are mammals.	

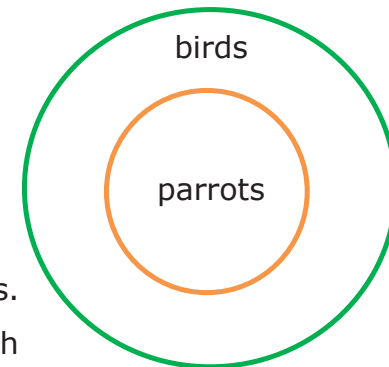
It may be helpful to understand "all" statements using Venn diagrams.



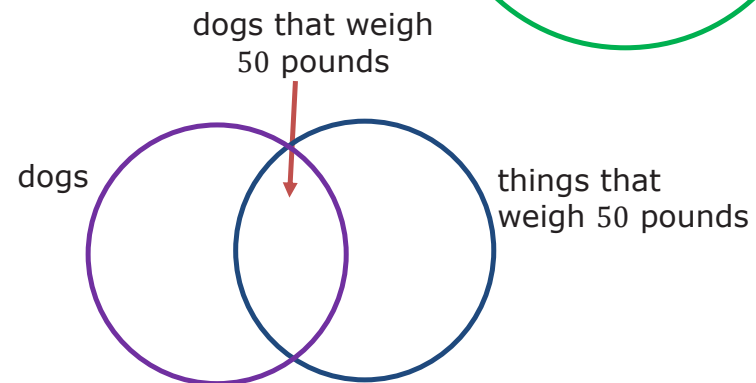
A **Venn diagram** is a diagram representing mathematical or logical sets pictorially as circles. Common elements of the sets are within the overlap of the circles.

7. The Venn diagram and a statement are shown.

All parrots are birds.

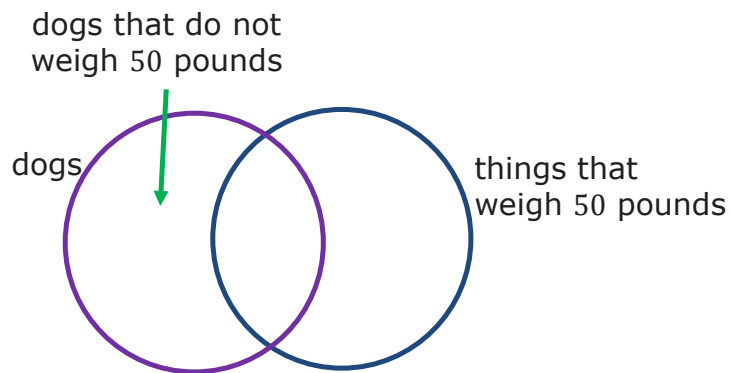


Some dogs weigh 50 pounds.



Discuss with your partner how the Venn diagrams help to show the difference between "all" and "some." Summarize your discussion.

8. The Venn diagram below shows where dogs that do not weigh 50 pounds would be placed.



Write a statement that matches the Venn diagram.



The **negation** of a statement is the opposite of the statement. If the statement is true, then the negation must be false.

9. Use the definition of negation to write the negation of each statement.
- Today is Wednesday.
 - Seo-jun is not wearing blue.

10. Rolando and Rhianna both wrote the negation of the statement, " $\sqrt{5}$ is an irrational number."

Rolando	Rhianna
$\sqrt{5}$ is not an irrational number.	$\sqrt{5}$ is a rational number.

- Discuss with your partner both negations. Write your observations and questions for the class.
- Based on the class discussion, which student has written a correct negation?



2.6.3 Guided Practice: "All" Statements and Negation

1. Determine if each statement is true.

Statement	True or False
All congruent angles have the same measure.	
All odd numbers are prime.	
All points that lie in a plane are collinear.	

2. Rewrite each false statement using "some."

3. Write each statement as an if-then statement.

Statement	If-Then
All congruent angles have the same measure.	
All odd numbers are prime.	
All points that lie on a plane are collinear.	

4. Write the negation of each statement.

Statement	Negation
$\triangle DTM$ is a right triangle.	
$m\angle TYW + m\angle WYK = 136^\circ$	
$\angle RWF$ and $\angle HYV$ are complementary angles.	



2.6.4 Wrap-Up: Reflection

- Circle the icon that best describes your understanding of each concept below.



I need help.



I got it.



I could help someone.

I can write the converse of a mathematical statement.			
I can write the inverse of a mathematical statement.			
I can write the contrapositive of a mathematical statement.			
I can determine if a statement is biconditional.			
I can write and interpret an "all" statement.			
I can write and interpret a negation.			

Unit 2, Lesson 7: Proofs – Part 1



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can relate deductive reasoning to completing proofs based on assumptions and given statements.
- ☐ I can recognize paragraph, flow-chart, and two column proofs.